Where Does the Gain in Dynamic Convolution Come From?

A controlled study of routing, capacity and width on two backbones*

Ankit Kumar

Mechatronics Department, University of Agder, Grimstad, Norway

24 September 2026; revised 17 October 2026

Abstract

Dynamic convolution replaces one kernel with a bank of K kernels mixed by input-dependent weights. Its gains are usually credited to input-adaptive kernel selection, but published comparisons set it against a network with K times fewer parameters, so routing and capacity are confounded. We separate them with a constant-mixture control, a mismatched-input control and a frozen random router, on ResNet-18 and MobileNetV2, CIFAR-100 and Tiny-ImageNet, across 264 training runs. Two results hold on both backbones. The kernel bank on its own adds nothing measurable: a uniformly mixed bank is never significantly different from a plain network, and is statistically equivalent to one at a ± 0.5 pp margin in 7 of ten conditions; the other three, all on MobileNetV2, are inconclusive. And where dynamic convolution helps, routing carries the effect, while no bank size beyond $K = 2$ is distinguishable from $K = 2$. Three ResNet-18 results do not transfer. On ResNet-18 a plain network of twice the width beats the routed one at matched parameters in all three pairs, by 2.3 to 4.3 pp; on MobileNetV2 the routed network is not distinguishable from a plain network with about twice its parameters in two of three pairs. On ResNet-18 the routing effect vanishes on CIFAR-100 at width 64; on MobileNetV2 it is +1.360 pp there. And on MobileNetV2, annealing the softmax temperature, which DY-Conv presents as necessary, costs about 1 pp at every width. MobileNetV2 adds a result with no ResNet-18 counterpart: a router frozen at its random initialisation is not distinguishable from a trained one and delivers the full benefit over the constant mixture, while routing each image with another image's descriptor removes it. Routing entropy, measured on ResNet-18, does not track the benefit. Of the four predictions we recorded before running the experiments that test them, three failed, and we report each one.

1 Introduction

A dynamic convolution layer stores K kernels and combines them per input. For an input feature map F and kernel bank $\{\mathcal{W}_k\}_{k=1}^{K}$, the layer computes

$$\mathcal{W}_{\text{dyn}}(F) = \sum_{k=1}^{K} z_k(F)\,\mathcal{W}_k, \qquad \mathbf{z} = \operatorname{softmax}\!\left(\frac{g(\phi(F))}{\tau}\right), \tag{1}$$

where ϕ summarises the feature map, g is a small multilayer perceptron and τ is a softmax temperature. Nearly every published variant sets ϕ to global average pooling (Yang et al., 2019; Chen et al., 2020).

The usual account of why this helps is that $\mathbf{z}$ adapts the kernel to the input. A second account is available and is rarely tested. The bank multiplies the stored convolution parameters by K, and published comparisons are made against a network with K times fewer parameters. Under that comparison, a gain from routing and a gain from extra capacity look identical.

This report separates the two. We compare the standard routed layer against a constant mixture that has the same bank and no routing, a router fed a different image's descriptor, a router frozen at random initialisation, and a plain network. The comparison spans three widths, three bank sizes, two temperature schedules, two datasets and two backbones: ResNet-18, where almost every parameter sits in a dynamic layer, and MobileNetV2, where the dynamic layers are depthwise and hold 3 % of the parameters. That is 264 training runs and 458 GPU-hours on one RTX 4070 Ti. Every comparison is paired by seed and tested for equivalence as well as for difference, so that a null result is a claim with a stated margin rather than a failure to reject. The MobileNetV2 program repeats the ResNet-18 program exactly, and its predictions were recorded, dated, before its first run.

Two findings hold on both backbones:

1. The kernel bank adds nothing measurable on its own (Sections 7 and 13.2).
2. Where dynamic convolution helps, the routing term carries the effect, and banks larger than $K = 2$ add nothing further (Sections 8 and 13.5).

Four depend on the backbone (Section 14):

*Draft of 17 October 2026. ResNet-18 and MobileNetV2 are complete; the same program on VGG-16, ResNet-50 and ConvNeXt-Tiny is pre-registered and running.

1. Whether width beats routing at equal parameters. It does on ResNet-18 in all three pairs we could form; on MobileNetV2 only at the smallest width (Sections 10 and 13.4).
2. Where routing helps. On ResNet-18 not on CIFAR-100 at width 64; on MobileNetV2 in all five conditions.
3. What temperature annealing does. On ResNet-18 it is neutral or harmful; on MobileNetV2 it is harmful at every width.
4. Whether the router has to be trained. On MobileNetV2 a router frozen at random initialisation does as well as a trained one (Section 13.3); on ResNet-18 the test is uninformative.

On ResNet-18, routing entropy does not measure whether routing is useful (Section 12).

We recorded four predictions before running the experiments that test them. On ResNet-18 we attributed the CIFAR-100 width-64 null to the network memorising its training set and predicted the effect would vanish on Tiny-ImageNet at width 64; it did not (Section 9). For MobileNetV2 we predicted that the bank term would be equivalent to zero in all five conditions, that width would beat routing in all three pairs, and that routing would help in at least three conditions. Only the last held (Section 13.6).

2 Related work

Dynamic convolution. CondConv (Yang et al., 2019) and DY-Conv (Chen et al., 2020) introduced input-dependent mixtures of kernels. ODConv (Li et al., 2022) extends the attention across several kernel dimensions, and KernelWarehouse (Li and Yao, 2024) reorganises the bank so that far larger K becomes affordable. The reported gains are large: for a ResNet-18 on ImageNet at four times the kernel parameters, Li and Yao (2024) report top-1 improvements of 3.38 pp for DY-Conv, 4.01 pp for ODConv and 5.61 pp for KernelWarehouse under their training recipe. FDConv (Chen et al., 2025) observes that the parallel kernels of DY-Conv tend toward highly similar frequency responses, so that the extra parameters buy little adaptability. SD-Conv (He et al., 2023) prunes half the parameters of a trained dynamic network at a cost below 1 pp on CIFAR-100, and Li et al. (2021) argue that the K static kernels are redundant. To our knowledge none of these works isolates the bank from the routing with a constant mixture of the same bank. Our finding that the bank contributes nothing on its own is consistent with these observations. Spatially resolved variants such as DRConv (Chen et al., 2021) route on local rather than global signals and fall outside the controls used here.

Optimisation difficulty. Li et al. (2021) identify the joint optimisation of dynamic attention and static kernels as the central difficulty of dynamic convolution and propose a matrix decomposition that eases it. DY-Conv addresses the same difficulty by annealing the softmax temperature from 30 to 1 early in training (Chen et al., 2020). Section 11 tests that schedule.

Re-parameterisation. A constant mixture of K kernels is a linear re-parameterisation of a single kernel. Such re-parameterisations can change optimisation without changing the function class (Arora et al., 2018), and ACNet and RepVGG exploit this by training a richer block and merging it into one kernel at inference (Ding et al., 2019, 2021). Our `uniform` arm is exactly such a mergeable block.

Memorisation. Networks of the size used here can fit their training sets completely, even under random labels (Zhang et al., 2017). That motivated the saturation hypothesis tested in Section 9.

Equivalence testing. The two one-sided tests procedure (Schuirmann, 1987) asserts that a difference lies within a stated margin; Lakens (2017) give a practical treatment. It is standard in bioequivalence studies and rare in this literature, where a large p-value from a difference test is often read as evidence of no effect.

3 Method

3.1 Routing signals

All arms share Eq. (1), the backbone, the optimiser and the seeds, and differ only in how $\mathbf{z}$ is produced. Write $\mu(F) \in \mathbb{R}^C$ for the channel-wise global average.

`static` No dynamic layer: the plain backbone, $K = 1$.

`uniform` $g \equiv \mathbf{0}$, so $\mathbf{z} = (1/K, \ldots, 1/K)$ for every input. The bank exists and is trained; nothing is routed.

`gap` $\mathbf{z} = \mathrm{softmax}(g(\mu(F))/\tau)$ with g trained. The standard layer.

shuffled As gap, but sample i is routed with the descriptor of sample $\pi(i)$ for a permutation π redrawn on every forward pass, at evaluation as well as training. The distribution of $\mathbf{z}$ is preserved and the pairing between image and kernel is destroyed.

frozen_random As gap, with every weight of g frozen at its random initialisation. Routing varies with the input but is never optimised.

3.2 The constant mixture is a single kernel

Proposition 1. *If $\mathbf{z} = (1/K, \ldots, 1/K)$ for every input, the layer computes $F * \bar{\mathcal{W}}$ with $\bar{\mathcal{W}} = \frac{1}{K}\sum_k \mathcal{W}_k$, and its effective parameter count equals that of one static kernel.*

Proof. The mixture in Eq. (1) is constant, so $\mathcal{W}_{\text{dyn}} = \bar{\mathcal{W}}$ for every F. □

At inference uniform is therefore a plain ResNet-18. At width 64 it stores 44.18M parameters and has an effective count of 11.22M. It spans the same function class as static, so any difference between the two is an effect of optimisation. That makes it the right control for the extra parameters of the bank.

3.3 Two exact implementations

Convolution is linear in its weights, so

$$F * \Big(\sum_k z_k \mathcal{W}_k \Big) = \sum_k z_k \, (F * \mathcal{W}_k). \tag{2}$$

The left side builds a per-sample kernel and runs one grouped convolution; at width 64 and batch 128 its aggregated weights alone take about 1.2 GB per stage-4 layer. The right side runs K ordinary convolutions and mixes the outputs, which costs K times the arithmetic but no per-sample weights. Every run uses the right side. The test suite asserts that the two agree to 10^{-10} in double precision across four routing signals.

3.4 Training protocol

Data. CIFAR-100 (Krizhevsky, 2009), 32×32, 100 classes, at widths 16, 32 and 64. Tiny-ImageNet (Le and Yang, 2015), 64×64, 200 classes, at widths 32 and 64. The stem is the same on both: a 3×3 stride-1 convolution with no max-pool. At 64×64 this leaves an 8×8 final feature map rather than 4×4 and roughly quadruples the cost of the deeper stages, but changing the stem would change the network between datasets. Crop padding is 4 pixels at 32 and 8 at 64.

Network and recipe. ResNet-18 (He et al., 2016) in which all 16 3×3 convolutions inside residual blocks are dynamic; the stem and the 1×1 shortcuts stay static. $K = 4$ unless stated. SGD with Nesterov momentum 0.9, learning rate 0.1, weight decay 5×10^{-4}, batch 128, five warm-up epochs then cosine decay over 150 epochs. Augmentation is random cropping and horizontal flipping only. Mixed precision in bfloat16. Six seeds, 0 to 5, shared by every arm. Batch 128 held at Tiny-ImageNet width 64 after a one-epoch memory probe, so the two Tiny-ImageNet widths are directly comparable.

Second backbone. MobileNetV2 (Sandler et al., 2018), adapted to small inputs in the usual way: the stem and the second stage use stride 1. Width 64 is the standard network; widths 32 and 16 scale every channel count by $1/2$ and $1/4$, rounded to a multiple of 8. The rule that decides which layers are dynamic is copied from ResNet-18 without change: every $k \times k$ convolution with $k > 1$ inside a block. In MobileNetV2 that is the 17 depthwise 3×3 convolutions; the stem, every 1×1 expansion and projection, and the classifier stay static. The consequences differ from ResNet-18. At CIFAR-100 width 64 the bank adds 0.19M parameters to a 2.35M network, so only 3 % of the static parameters sit in dynamic layers, and the router of the gap arm (the ResNet-18 design, a two-layer perceptron of hidden width $C_{\text{in}}/4$) has 1137K parameters, 48 % of the static network and more than the bank it routes. Static convolutions and every bank kernel share one initialisation, Kaiming-normal with fan-out. Recipe, arms, seeds and margin are those of ResNet-18; every group ran at batch 128.

Temperature. $\tau = 1$ throughout, except in Section 11, where τ falls linearly from 30 to 1 over the first ten epochs and stays at 1 thereafter.

4 Statistical protocol

Equivalence. Several conclusions are null conclusions, and a large p from a difference test does not establish one. For paired differences d_i with mean $\bar{d}$ and standard error $s_{\bar{d}}$ we use two one-sided tests against a margin δ,

$$p_{\mathrm{TOST}} = \max\left\{1 - T_{n-1}\left(\frac{\bar{d}+\delta}{s_{\bar{d}}}\right),\ T_{n-1}\left(\frac{\bar{d}-\delta}{s_{\bar{d}}}\right)\right\}, \tag{3}$$

and read $p_{\mathrm{TOST}} < 0.05$ as a positive claim that the true difference lies within $\pm\delta$. We fixed $\delta = 0.5$ pp before the first run. The pre-registration justified this by reference to reported gains at CIFAR scale without citing a source. The nearest documented figures we have since found are the ImageNet ResNet-18 gains of 3.38 to 5.61 pp quoted above, and the margin sits an order of magnitude below them.

Why six seeds. The two-sided Wilcoxon signed-rank test (Wilcoxon, 1945) on n pairs admits only 2^n equally likely sign assignments, so its smallest attainable p is 2^{1-n}: 0.25 at $n = 3$, 0.0625 at $n = 5$, 0.031 at $n = 6$. Six is the smallest count at which it can reach 0.05. We report it beside the paired t-test throughout.

Power. For `gap` − `uniform` on CIFAR-100 at width 64 the paired standard deviation is 0.257 pp, giving a minimum detectable difference of 0.270 pp at $n = 6$. Simulated power (2×10^5 draws) is 48.5 % for a true effect of 0.25 pp, 96.3 % at 0.5 pp and 100.0 % at 1.0 pp. The width-64 null on CIFAR-100 is not a power artefact.

Pre-registration. Arms, recipe, primary metric, seeds, margin and the decision table were written down before the first run finished. The primary metric is final-epoch top-1 accuracy, since best-epoch accuracy maximises over a noisy sequence. The Tiny-ImageNet width-64 prediction of Section 9 was recorded in writing, dated, before any of those runs. An expectation about Tiny-ImageNet width 32 was stated before those runs but not recorded; we give it correspondingly less weight. For MobileNetV2, three predictions and a gate (the first static run had to reach 65 % on CIFAR-100 before the other 131 runs were spent) were recorded on 30 September 2026, before any MobileNetV2 run (Section 13.6).

5 ResNet-18 on CIFAR-100 at the standard width

Sections 5 to 12 report ResNet-18. Section 13 reports MobileNetV2 and Section 14 compares the two.

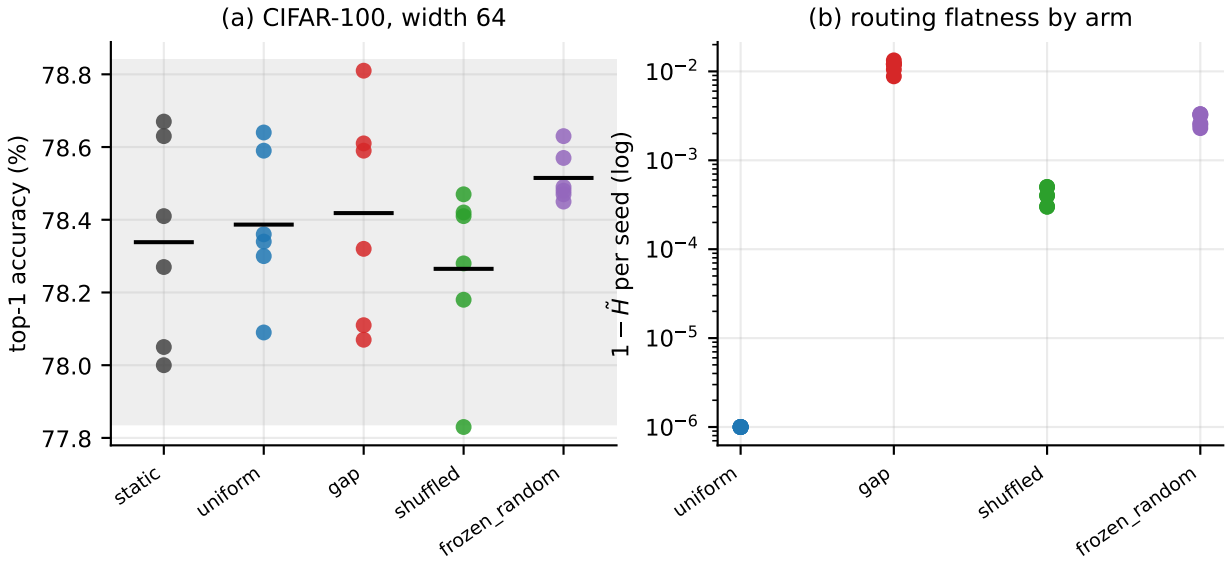


Figure 1: CIFAR-100, width 64. (a) Every run of the five arms; bars are arm means and the shaded band is ±0.5 pp around `static`. (b) Router flatness $1 - \tilde{H}$ per seed, log scale; `uniform` is exactly zero and is drawn at the axis floor.

At width 64 every arm is equivalent to a plain ResNet-18 (Table 18, Figure 1). The largest gap between arm means is 0.250 pp, against a median within-arm seed standard deviation of 0.238 pp. A one-way ANOVA across the five arms gives $F(4, 25) = 0.954$, $p = 0.450$.

A router whose weights were never trained matches a trained one (`frozen_random` − `gap` = +0.097 pp, $p = 0.443$, $p_{\mathrm{TOST}} = 0.0089$), and so does a router that routes each image with another image's descriptor. The trained router is not degenerate: its per-seed mean entropy stays below that of the controls (Figure 1b). It discriminates between kernels, and at this width the discrimination changes nothing.

6 ResNet-18: the capacity sweep on CIFAR-100

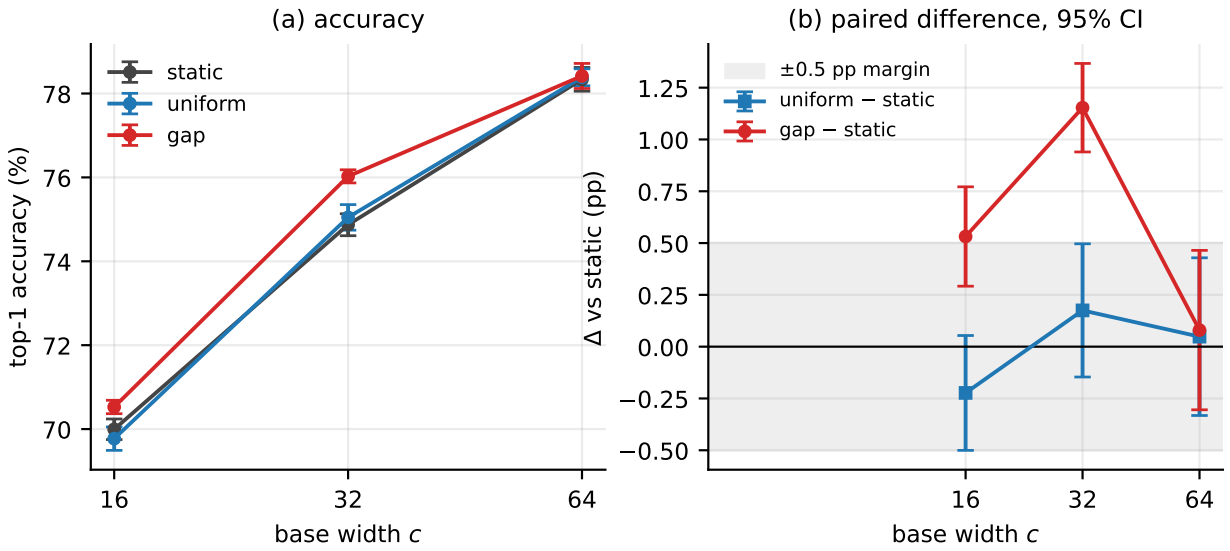


Figure 2: CIFAR-100. (a) Accuracy against base width, ± 1 standard deviation over six seeds. (b) Paired difference from `static` with 95 % confidence intervals.

Reducing width changes the answer (Figure 2). `gap` − `static` is +0.532 pp at width 16 ($p = 0.0023$), +1.153 pp at width 32 ($p < 10^{-4}$) and +0.080 pp at width 64 ($p = 0.6158$). The effect is not monotone in capacity: it peaks at width 32. MobileNetV2 shows no such peak and no null at width 64 (Section 13.2). `uniform` is equivalent to `static` at all three widths. Figure 16 in the appendix shows every run.

7 ResNet-18: the kernel bank contributes nothing

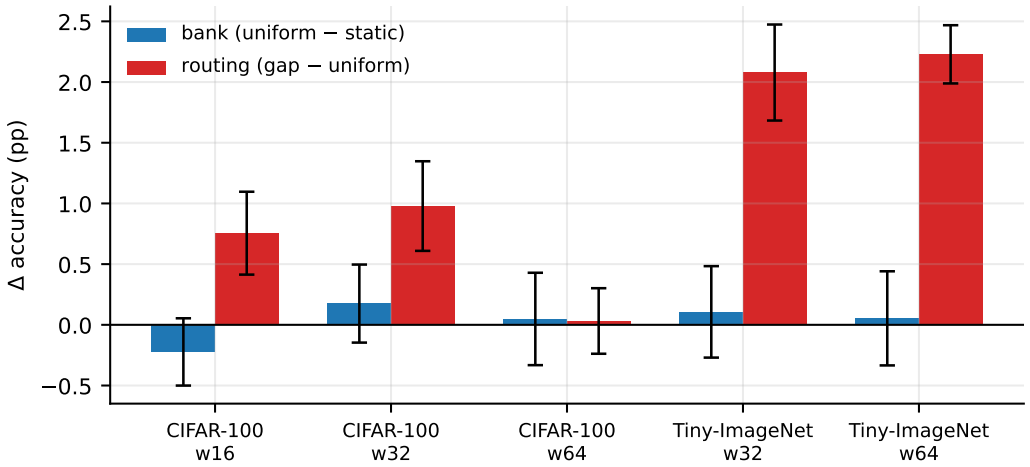


Figure 3: Decomposition of the effect of dynamic convolution into the bank (`uniform` − `static`) and routing (`gap` − `uniform`), five conditions, 95 % CIs.

Splitting the total effect `gap` − `static` into a bank term and a routing term separates the two accounts of the introduction. Table 1 and Figure 3 give the split in all five conditions that have a `uniform` arm.

The bank term is equivalent to zero in every one of them, on both datasets, with p_{TOST} between 0.014 and 0.025. The largest bank term observed is +0.175 pp. Since the bank contributes nothing within the margin, the total effect is the routing effect: on Tiny-ImageNet at width 64, for example, routing is +2.228 pp of a +2.282 pp total. At CIFAR-100 width 16 the bank term is slightly negative (-0.223 pp, equivalent) and the routing term therefore exceeds the total, which is why we report no percentage share.

Table 1: Bank and routing terms, paired by seed. The bank term is equivalent to zero in all five conditions.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	-0.223	0.0251	equivalent	+0.755	0.0023	+0.532	0.0023
CIFAR-100	32	+0.175	0.0242	equivalent	+0.978	0.0010	+1.153	$< 10^{-4}$
CIFAR-100	64	+0.048	0.0142	equivalent	+0.032	0.7752	+0.080	0.6158
Tiny-ImageNet	32	+0.107	0.0218	equivalent	+2.078	$< 10^{-4}$	+2.185	$< 10^{-4}$
Tiny-ImageNet	64	+0.053	0.0157	equivalent	+2.228	$< 10^{-4}$	+2.282	$< 10^{-4}$

8 ResNet-18: bank size

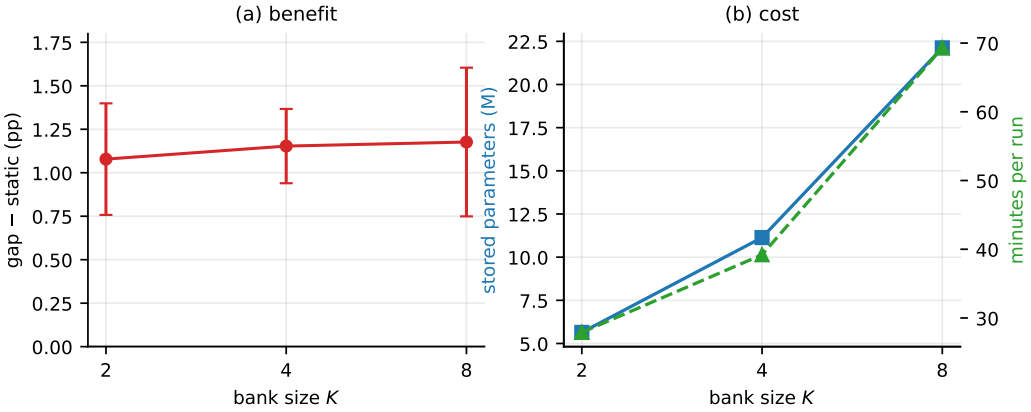


Figure 4: CIFAR-100, width 32. (a) Effect over `static` for $K \in \{2, 4, 8\}$, 95 % CIs. (b) Stored parameters and training time.

The width-32 peak was measured at $K = 4$; Table 2 repeats it at $K = 2$ and $K = 8$. All three bank sizes beat `static` by a similar margin, and no pair of them is distinguishable (Table 3): $K = 8$ over $K = 2$ is +0.098 pp ($p = 0.409$). The benefit is already complete at $K = 2$. Going from $K = 2$ to $K = 8$ multiplies stored parameters by 3.9 and training time by 2.5 (Figure 4).

Table 2: Bank size at CIFAR-100 width 32. Differences are against the `static` row.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	75.950	0.148	+1.078	[+0.758, +1.399]	0.0003	5.64	27.9
4	76.025	0.156	+1.153	[+0.940, +1.367]	$< 10^{-4}$	11.13	39.2
8	76.048	0.242	+1.177	[+0.749, +1.604]	0.0009	22.12	69.3
`static`	74.872	0.262	–	–	–	2.82	8.6

Table 3: Pairwise comparisons between bank sizes.

comparison	Δ (pp)	95 % CI	p	p_{TOST}	verdict
$K = 4$ vs $K = 2$	+0.075	[-0.087, +0.237]	0.2878	0.0005	equivalent
$K = 8$ vs $K = 4$	+0.023	[-0.286, +0.332]	0.8538	0.0053	equivalent
$K = 8$ vs $K = 2$	+0.098	[-0.182, +0.379]	0.4091	0.0072	equivalent

9 ResNet-18 on Tiny-ImageNet, and a failed prediction

9.1 Width 32

On Tiny-ImageNet at width 32, `gap − static` is +2.185 pp (95 % CI [+1.781, +2.589], $p < 10^{-4}$), against +1.153 pp on CIFAR-100 at the same width. The bank term is +0.107 pp and equivalent to zero. Before these runs we expected the

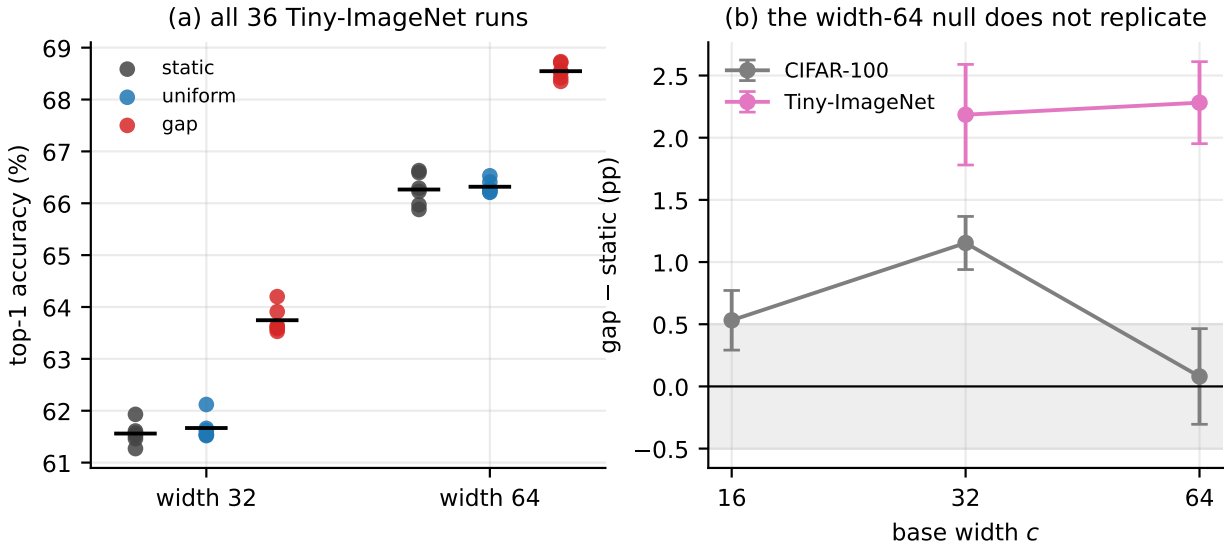


Figure 5: (a) All 36 Tiny-ImageNet runs. (b) `gap − static` by width on both datasets, 95 % CIs. The CIFAR-100 width-64 null does not appear on Tiny-ImageNet.

effect to be larger than on CIFAR-100, since the static arm trains to a higher loss (0.1194 against 0.0170) and so sits further from memorisation. That expectation held, but it was stated and not recorded, it predicted only a direction, and we do not treat it as strong evidence.

9.2 Width 64: the recorded prediction

The saturation hypothesis says the routing effect vanishes once a network memorises its training data. On CIFAR-100 the width-64 null coincides with a final training loss of 0.0080. Before running Tiny-ImageNet at width 64 we recorded, in writing and dated, the prediction in Table 4: that the effect would fall below 0.5 pp and be equivalent to zero, and that the static arm would train to a loss below 0.05.

Table 4: The prediction recorded before the Tiny-ImageNet width-64 runs, and the outcome. Both conditions were stated as required.

	prediction	measured	outcome
(a)	`gap − static` below 0.5 pp, equivalent by TOST	+2.282 pp, $p = 1.0 \times 10^{-5}$, $p_{\mathrm{TOST}} = 1.0000$	failed
(b)	static final training loss below 0.05	0.021	held

The loss condition held (0.0210) and the effect did not vanish: `gap − static` $= +2.282$ pp, 95 % CI [+1.952, +2.611], $p = 1.0 \times 10^{-5}$. The prediction failed. The bank term is again equivalent to zero (+0.053 pp, $p_{\mathrm{TOST}} = 0.0157$), so the effect is carried by routing, and it is as large at width 64 as at width 32.

9.3 What remains of the saturation account

Table 5 orders the five conditions by the final training loss of the static arm. The effects do not follow that order (Figure 6b). Tiny-ImageNet at width 64 trains to a loss close to CIFAR-100 at width 32 and shows twice the effect; CIFAR-100 at width 16 trains to the highest loss and shows one of the smallest. A threshold near 0.01 would separate the one null from the four positive results, but it would be chosen after seeing the data, and we do not adopt it.

What survives is narrower. On CIFAR-100 the routing effect vanishes at width 64, and that result is well powered and holds under two temperature schedules. It does not replicate on Tiny-ImageNet, and training loss does not explain it. Plausible differences between the datasets include the input resolution, the final feature-map size and the number of classes; none of them was isolated here.

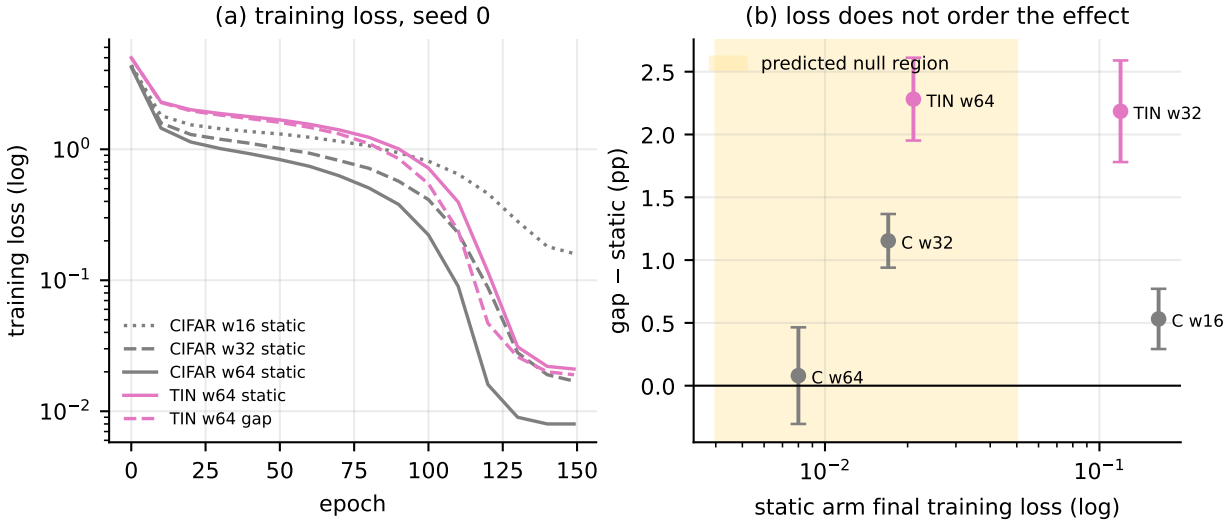


Figure 6: (a) Training loss of static and gap arms, seed 0, log scale. (b) Routing effect against the final training loss of the static arm in all five conditions. The shaded band is the region the prediction associated with a null.

Table 5: Five conditions ordered by the final training loss of the static arm.

dataset	width	static train loss	gap − static (pp)	95 % CI	verdict
CIFAR-100	64	0.0080	+0.080	[-0.305, +0.465]	equivalent
CIFAR-100	32	0.0170	+1.153	[+0.940, +1.367]	higher
Tiny-ImageNet	64	0.0210	+2.282	[+1.952, +2.611]	higher
Tiny-ImageNet	32	0.1194	+2.185	[+1.781, +2.589]	higher
CIFAR-100	16	0.1643	+0.532	[+0.292, +0.771]	higher

10 ResNet-18: matched parameter budgets

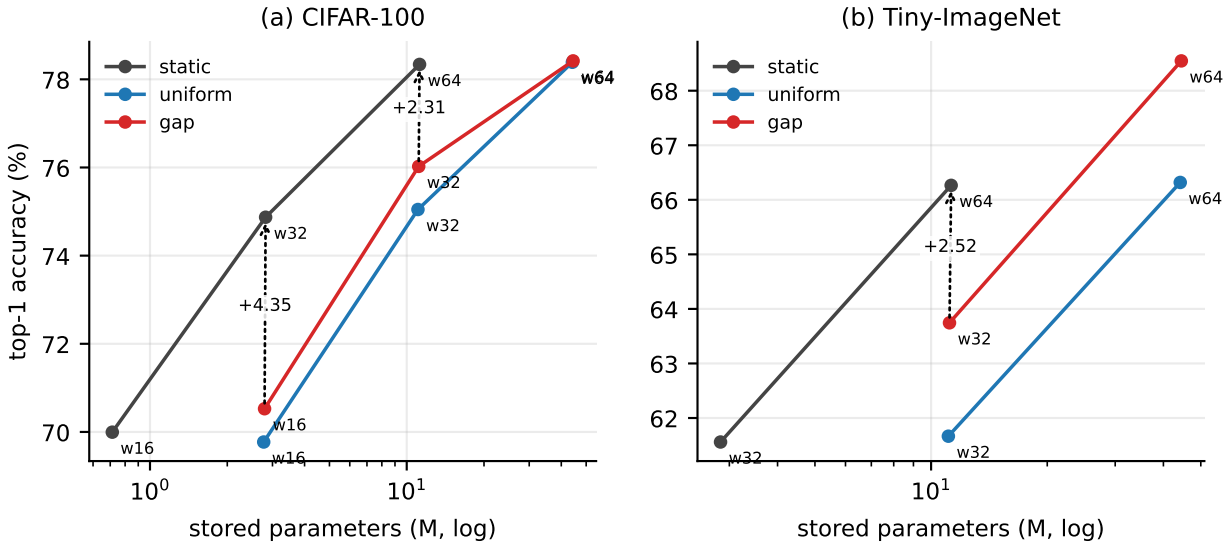


Figure 7: Accuracy against stored parameters. Dotted arrows join routed and plain networks matched to within 1 % on parameter count.

Because widths differ by a factor of two and the bank multiplies convolution parameters by four, a routed network at one width stores almost exactly as many parameters as a plain network at the next width up. That gives three matched comparisons without any extra runs (Table 6, Figure 7).

The plain network wins all three: by +4.345 pp at 2.8M parameters and +2.313 pp at 11.2M on CIFAR-100, and by +2.520 pp at 11.2M on Tiny-ImageNet. It also trains faster in each case; on Tiny-ImageNet, 129.0 min per run

against 284.2 min. A fourth, unmatched comparison points the same way: the $K = 8$ network at width 32 stores 22.1M parameters, twice the plain width-64 network, and trails it by 2.29 pp. This result depends on almost every parameter of ResNet-18 sitting in a dynamic layer, and it does not transfer to MobileNetV2 (Section 13.4).

Table 6: Routed against plain networks at matched parameter counts, paired by seed. Minutes are routed / plain per run. The last column is the routed network's stored parameters divided by the plain one's; a pair is called matched only within 10 %.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	min/run	param ratio
CIFAR-100	`gap` w16 (2.79M)	`static` w32 (2.82M)	70.527	74.872	+4.345	[+4.103, +4.587]	$< 10^{-4}$	33.7 / 8.6	0.99 (matched)
CIFAR-100	`gap` w32 (11.13M)	`static` w64 (11.22M)	76.025	78.338	+2.313	[+1.870, +2.757]	$< 10^{-4}$	39.2 / 18.8	0.99 (matched)
Tiny-ImageNet	`gap` w32 (11.16M)	`static` w64 (11.27M)	63.745	66.265	+2.520	[+2.311, +2.729]	$< 10^{-4}$	284.2 / 129.0	0.99 (matched)

11 ResNet-18: temperature schedule

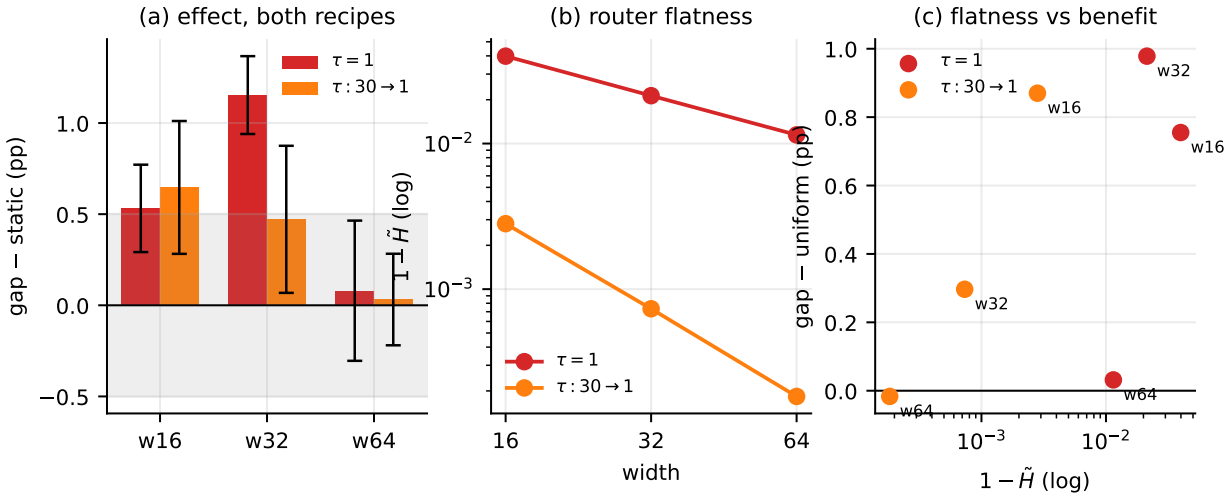


Figure 8: CIFAR-100. (a) Routing effect under a constant and an annealed temperature, 95 % CIs. (b) Router flatness. (c) Flatness against benefit over `uniform`; the two schedules do not lie on one curve.

With $\tau = 1$ the softmax can be close to one-hot from the first step, starving most of the bank of gradient. DY-Conv anneals τ from 30 to 1 for this reason (Chen et al., 2020). If the CIFAR-100 width-64 null came from an undertrained bank, it would say nothing about dynamic convolution, so we repeated `gap` at all three widths with the annealed schedule (Table 7).

The width-64 null survives: annealed `gap` against `static` is +0.032 pp with $p_{\text{TOST}} = 0.0025$, a tighter equivalence than the 0.0188 obtained at $\tau = 1$. Annealing is otherwise neutral at width 64 (-0.048 pp, $p = 0.7449$) and width 16 (+0.115 pp, $p = 0.4313$), and harmful at width 32 (-0.682 pp, $p = 0.0017$), where it also leaves a higher final training loss (0.0173 against 0.0132). Backbone, dataset, bank size and schedule length all differ from the DY-Conv setting, so this is not evidence against their result; it is evidence that the CIFAR-100 null does not depend on the choice of schedule. On MobileNetV2 annealing is harmful at all three widths (Section 13.5).

At width 64 the annealed router went flat during the high-τ phase and stayed flat for the remaining 140 epochs: it is 63 times flatter than the constant router ($1 - \tilde{H} = 0.00018$) and equivalent to `uniform` (-0.017 pp, $p_{\text{TOST}} = 0.0007$). This follows from Proposition 1. A flat mixture trains the bank toward its own mean, and a homogeneous bank leaves the router nothing to discriminate between. At width 16 the same schedule did not produce that outcome, which the next section takes up.

Table 7: Temperature ablation on CIFAR-100, `gap` arm. The last two columns test whether the annealed router behaves like the constant mixture.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p_{TOST}	Δ	verdict
16	70.527	70.642	+0.115	0.4313	+0.532	+0.647	0.8259	+0.870	higher
32	76.025	75.343	-0.682	0.0017	+1.153	+0.472	0.4319	+0.297	inconclusive
64	78.418	78.370	-0.048	0.7449	+0.080	+0.032	0.0025	-0.017	equivalent

12 ResNet-18: routing entropy does not measure routing usefulness

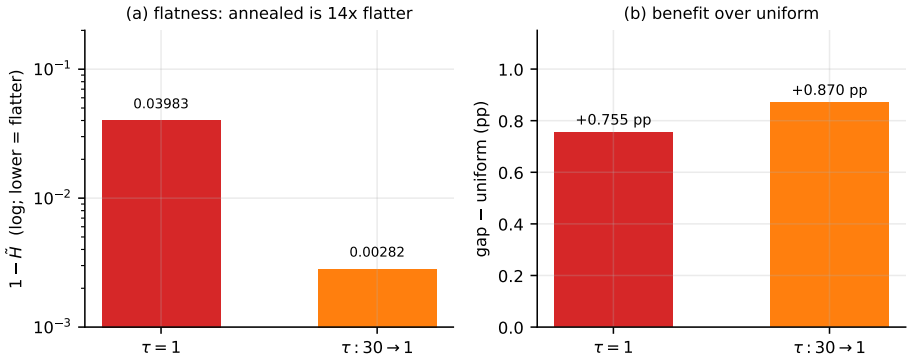


Figure 9: CIFAR-100, width 16. The annealed router is 14 times flatter by entropy and delivers the larger benefit over `uniform`.

Normalised routing entropy $\tilde{H} = H(\mathbf{z})/\log K$ is the statistic this literature reports to show that a router is doing something. Table 8 sets it against the benefit each router delivers over `uniform` in six conditions.

At width 16 the two schedules give a counterexample (Figure 9). The constant router has $1 - \tilde{H} = 0.03983$ and beats `uniform` by +0.755 pp. The annealed router has $1 - \tilde{H} = 0.00282$, 14 times flatter, and beats `uniform` by +0.870 pp ($p = 0.0044$). A router that looks nearly inactive on the standard diagnostic delivered more than one that looks clearly active.

The reason is that entropy compresses the signal quadratically. Writing $z_k = 1/K + \varepsilon_k$ with $\sum_k \varepsilon_k = 0$ and expanding to second order,

$$1 - \tilde{H} \;\approx\; \frac{K}{2 \log K} \sum_k \varepsilon_k^2, \tag{4}$$

so the deviations that carry input-dependent information enter squared. At $1 - \tilde{H} = 0.00282$ the root-mean-square deviation from $1/K$ is about 0.022, so typical mixing weights lie in [0.228, 0.272], roughly 9 % of $1/K$. Entropy reports that as 0.997.

Entropy is not meaningless. The annealed router at width 64, at $1 - \tilde{H} = 0.00018$, has collapsed in the sense that matters: it is equivalent to `uniform`. But across the six conditions of Table 8, sorting by $1 - \tilde{H}$ does not sort the benefit, and the two schedules do not lie on one curve (Figure 8c). A value of $\tilde{H}$ below one is not evidence that routing is useful, and a value near one is not evidence that it is not.

Table 8: Routing entropy against the benefit it is taken to indicate, CIFAR-100. RMS dev is the root-mean-square deviation of the mixing weights from $1/K$, recovered from Eq. (4).

width	τ	$\tilde{H}$	$1 - \tilde{H}$	RMS dev	`gap`−`uniform` (pp)	p
16	1	0.96017	0.03983	0.0831	+0.755	0.0023
16	$30 \to 1$	0.99718	0.00282	0.0221	+0.870	0.0044
32	1	0.97868	0.02132	0.0608	+0.978	0.0010
32	$30 \to 1$	0.99927	0.00073	0.0113	+0.297	0.0810
64	1	0.98853	0.01147	0.0446	+0.032	0.7752
64	$30 \to 1$	0.99982	0.00018	0.0056	-0.017	0.8369

13 A second backbone: MobileNetV2

13.1 Setup and coverage

The program of Sections 5 to 12 was repeated on MobileNetV2 with no change to arms, seeds, recipe, margin or the rule that decides which layers are dynamic (Section 3). The gate run reached 72.66 % against the recorded floor of 65 %, and all 132 runs finished in 254 GPU-hours. None diverged and every group trained at batch 128. One run (Tiny-ImageNet, width 64, `gap`, seed 1, final accuracy 61.15 %) was stopped by a GPU fault at epoch 125 and resumed from its checkpoint; without it, `gap − static` in that condition is +1.526 pp ($p = 0.0007$, $n = 5$), so no conclusion depends on it. The MobileNetV2 numbers in this draft are taken from the training log, which prints every run's final accuracy and training loss; the routing entropies are recorded only in the result files and are not analysed here.

13.2 The bank adds nothing measurable; routing helps everywhere

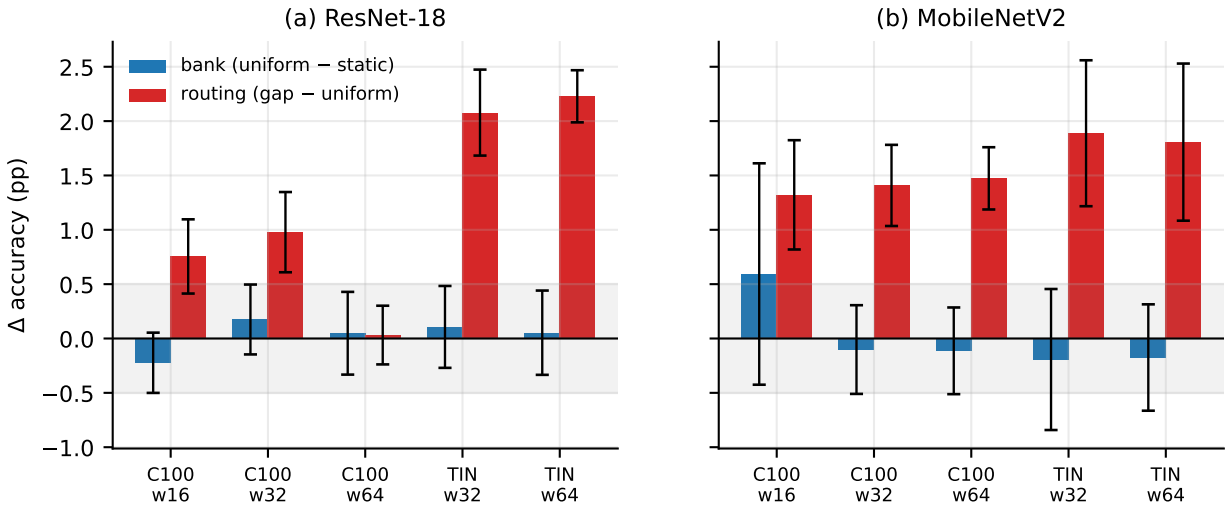


Figure 10: Bank term (`uniform − static`) and routing term (`gap − uniform`) in the five conditions, for (a) ResNet-18 and (b) MobileNetV2, 95 % CIs. The shaded band is the ±0.5 pp margin.

On MobileNetV2 the bank term is never significantly different from zero (Table 9, Figure 10). It is equivalent to zero on CIFAR-100 at widths 32 and 64 ($p_{\text{TOST}} = 0.0269$ and 0.0275). At CIFAR-100 width 16 and at both Tiny-ImageNet widths it is inconclusive: the point estimates (+0.593, -0.193 and -0.175 pp) lie inside the margin, but the confidence intervals do not. At CIFAR-100 width 16 the static arm itself is unusually variable across seeds (Table 22).

The routing term is positive with $p < 0.05$ in all five conditions, from 1.32 to 1.89 pp, and the total effect ranges from 1.31 to 1.91 pp. Unlike ResNet-18, the effect does not vanish on CIFAR-100 at width 64: `gap − static` is +1.360 pp there (95 % CI [+0.986, +1.734], $p = 0.0002$), and the effect is roughly flat across width on both datasets (Figure 11).

Table 9: MobileNetV2: bank and routing terms, paired by seed.

		bank (`uniform−static`)			routing (`gap−uniform`)		total	
dataset	width	Δ (pp)	p_{TOST}	verdict	Δ (pp)	p	Δ (pp)	p
CIFAR-100	16	+0.593	0.5884	inconclusive	+1.322	0.0011	+1.915	0.0225
CIFAR-100	32	-0.102	0.0269	equivalent	+1.408	0.0002	+1.307	0.0015
CIFAR-100	64	-0.113	0.0275	equivalent	+1.473	$< 10^{-4}$	+1.360	0.0002
Tiny-ImageNet	32	-0.193	0.1393	inconclusive	+1.888	0.0008	+1.695	0.0028
Tiny-ImageNet	64	-0.175	0.0742	inconclusive	+1.807	0.0014	+1.632	0.0002

13.3 An untrained router does as well as a trained one

On CIFAR-100 at width 64 the five MobileNetV2 arms differ (one-way ANOVA $F = 42.8$, $p < 10^{-4}$), where the ResNet-18 arms did not ($F = 0.954$, $p = 0.450$). The controls separate cleanly (Table 10, Figure 12).

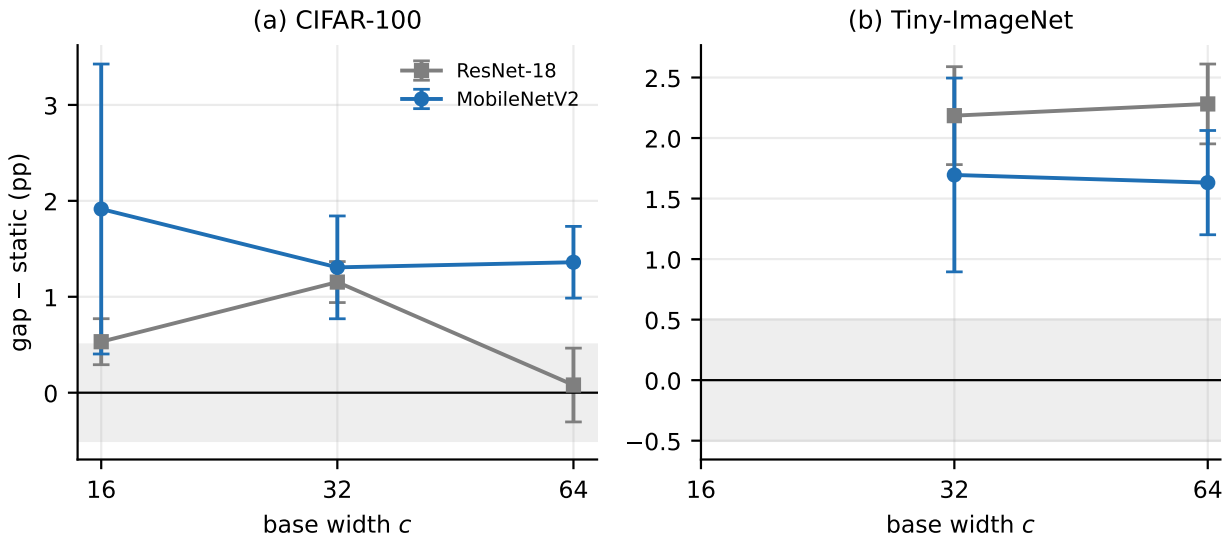


Figure 11: `gap − static` against width for both backbones, 95 % CIs. ResNet-18 has a null on CIFAR-100 at width 64; MobileNetV2 does not.

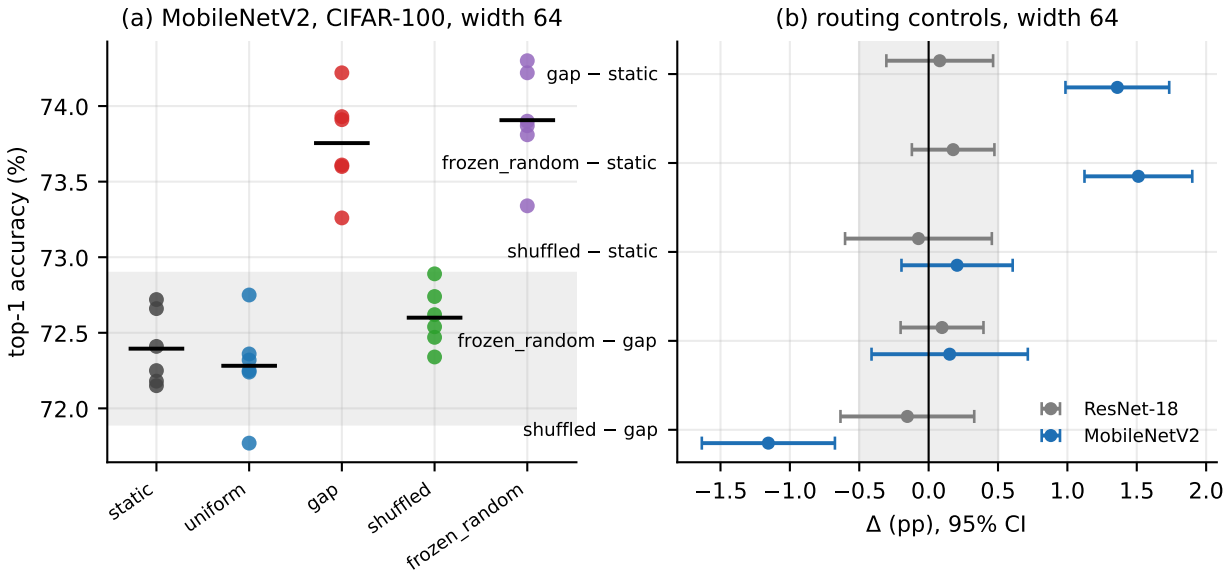


Figure 12: CIFAR-100, width 64. (a) Every MobileNetV2 run of the five arms; bars are arm means, the band is ±0.5 pp around `static`. (b) The routing controls on both backbones, 95 % CIs.

A router frozen at its random initialisation beats `static` by +1.512 pp and `uniform` by +1.625 pp ($p = 0.0008$), and is not distinguishable from the trained router (`frozen_random` − `gap` = +0.152 pp, 95 % CI [-0.411, +0.715], $p = 0.5196$). It is not equivalent to it at our margin either ($p_{\text{TOST}} = 0.0864$), so the claim is that no difference was detected, not that none exists. Routing each image with another image's descriptor removes the benefit: `shuffled` is not distinguishable from `static` (+0.205 pp, $p = 0.2456$) and falls below `gap` by -1.155 pp ($p = 0.0016$).

Read together, the benefit on MobileNetV2 requires that each image's own descriptor determine its mixture, and does not require the router to be trained. A fixed random map from descriptors to mixtures is enough; the bank learns kernels that suit the routing it is given. The frozen router's weights are held as buffers and are not trained, so the gain is not a product of the 1137K trainable router parameters either. On ResNet-18 the same controls are uninformative, since no arm differed from `static` at this width. The frozen router was run only on CIFAR-100 at width 64, and the claim is limited to that condition.

Table 10: Routing controls on CIFAR-100 at width 64. The ResNet-18 columns are repeated from Section 5 for comparison.

	ResNet-18		MobileNetV2				
comparison	Δ (pp)	verdict	Δ (pp)	95 % CI	p	p_{TOST}	verdict
`uniform` − `static`	+0.048	equivalent	-0.113	[-0.512, +0.285]	0.4978	0.0275	equivalent
`gap` − `static`	+0.080	equivalent	+1.360	[+0.986, +1.734]	0.0002	0.9990	higher
`shuffled` − `static`	-0.073	equivalent	+0.205	[-0.196, +0.606]	0.2456	0.0585	inconclusive
`frozen_random` − `static`	+0.177	equivalent	+1.512	[+1.123, +1.900]	0.0002	0.9994	higher
`frozen_random` − `gap`	+0.097	equivalent	+0.152	[-0.411, +0.715]	0.5196	0.0864	inconclusive
`shuffled` − `gap`	-0.153	inconclusive	-1.155	[-1.635, -0.675]	0.0016	0.9915	lower
`frozen_random` − `uniform`	+0.128	equivalent	+1.625	[+1.051, +2.199]	0.0008	0.9980	higher

13.4 Parameter budgets

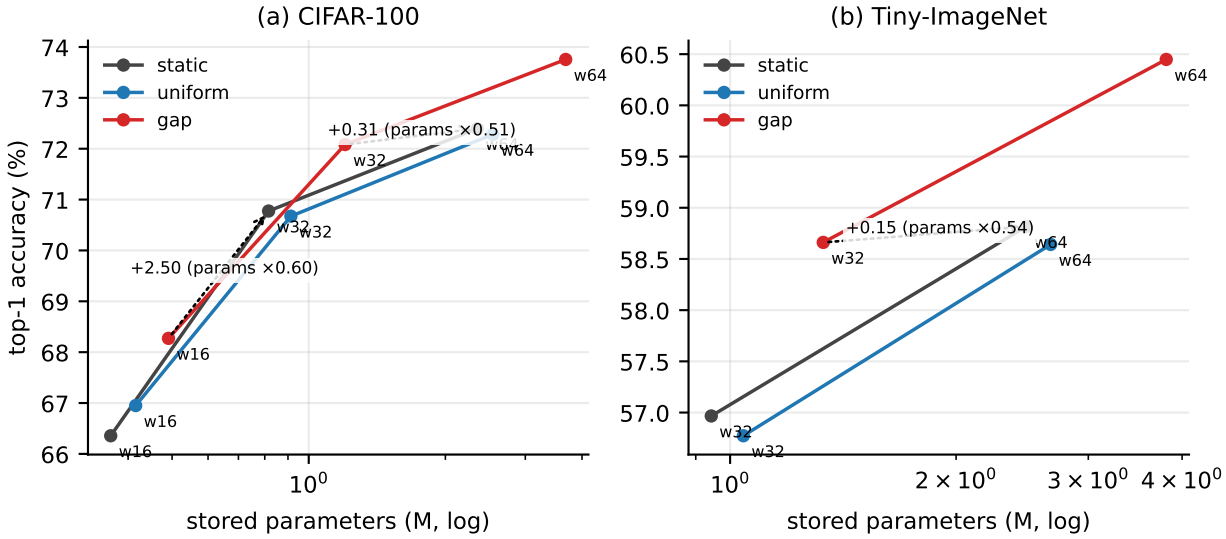


Figure 13: MobileNetV2: accuracy against stored parameters. Arrows join `gap` at width c to `static` at width $2c$; labels give the accuracy difference and the parameter ratio.

On ResNet-18, `gap` at width c and `static` at width $2c$ store nearly the same number of parameters. On MobileNetV2 they do not: because so few parameters sit in dynamic layers, the routed network has 0.60 to 0.51 times the parameters of the plain one (Table 11, Figure 13). Even so, the plain network with roughly twice the parameters wins only at the smallest width (+2.503 pp, $p = 0.0001$). On CIFAR-100 at width 32 and on Tiny-ImageNet the two are not distinguishable (+0.313 pp, $p = 0.1111$; +0.155 pp, $p = 0.5806$). Interpolating the static accuracy curve in log-parameters to the routed network's parameter count puts the routed network ahead in all three cases (static minus routed: -0.298, -0.713 and -1.037 pp); that comparison is an interpolation and has no confidence interval.

Table 11: MobileNetV2: `gap` at width c against `static` at width $2c$, paired by seed. Δ is plain minus routed. The ratio is the routed network's stored parameters over the plain one's; none of these pairs is matched. The last column compares `gap` with the static curve interpolated to its parameter count.

dataset	routed	plain	routed	plain	Δ (pp)	95 % CI	p	verdict	ratio	Δ equal params
CIFAR-100	`gap` w16 (0.49M)	`static` w32 (0.82M)	68.272	70.775	+2.503	[+1.927, +3.080]	0.0001	higher	0.60	-0.298
CIFAR-100	`gap` w32 (1.20M)	`static` w64 (2.35M)	72.082	72.395	+0.313	[-0.103, +0.730]	0.1111	inconclusive	0.51	-0.713
Tiny-ImageNet	`gap` w32 (1.33M)	`static` w64 (2.48M)	58.662	58.817	+0.155	[-0.520, +0.830]	0.5806	inconclusive	0.54	-1.037

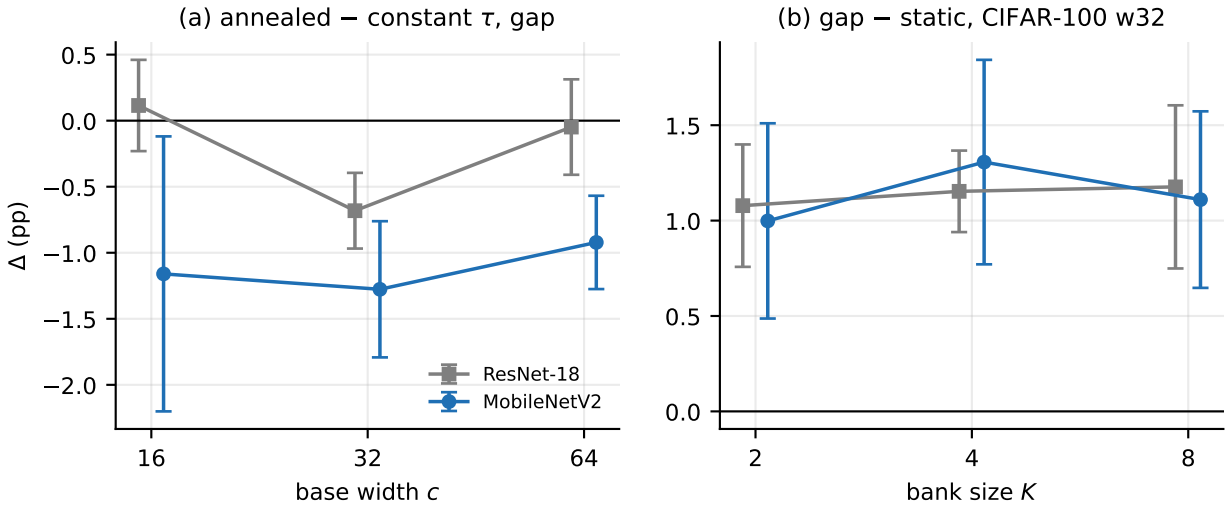


Figure 14: Both backbones, CIFAR-100. (a) Annealed minus constant temperature for `gap`. (b) `gap` − `static` at width 32 for $K \in \{2, 4, 8\}$. 95 % CIs.

13.5 Bank size and temperature

Bank size behaves as on ResNet-18 (Table 12, Figure 14b): all three sizes beat `static`, and $K = 8$ over $K = 2$ is +0.112 pp ($p = 0.5929$, inconclusive).

Temperature does not (Table 13, Figure 14a). Annealing τ from 30 to 1 lowers `gap` at every width: -1.160 pp at width 16 ($p = 0.0352$), -1.277 pp at width 32 ($p = 0.0014$) and -0.922 pp at width 64 ($p = 0.0011$). At width 32 the annealed router is equivalent to the constant mixture (+0.132 pp); at width 64 it remains above it (+0.552 pp). The annealed runs also end at a higher training loss at every width (0.713, 0.411, 0.225 against 0.655, 0.344, 0.168). On ResNet-18 at width 64 the annealed router went flat and stayed flat (Section 11); whether the same happens on MobileNetV2 needs the routing entropies, which this draft does not include.

Table 12: MobileNetV2: bank size at CIFAR-100 width 32.

K	accuracy	sd	Δ (pp)	95 % CI	p	params (M)	min/run
2	71.773	0.490	+0.998	[+0.487, +1.510]	0.0041	1.14	51.6
4	72.082	0.356	+1.307	[+0.771, +1.843]	0.0015	1.20	47.3
8	71.885	0.400	+1.110	[+0.647, +1.573]	0.0016	1.34	98.8
`static`	70.775	0.217	–	–	–	0.82	18.0

Table 13: MobileNetV2: temperature ablation on CIFAR-100, `gap` arm.

	accuracy		annealed − constant		vs `static`			annealed vs `uniform`	
width	$\tau = 1$	annealed	Δ	p	$\tau = 1$	annealed	p	Δ	verdict
16	68.272	67.112	-1.160	0.0352	+1.915	+0.755	0.3286	+0.162	inconclusive
32	72.082	70.805	-1.277	0.0014	+1.307	+0.030	0.8534	+0.132	equivalent
64	73.755	72.833	-0.922	0.0011	+1.360	+0.438	0.0051	+0.552	higher

13.6 The pre-registered predictions

Table 14 gives the three predictions recorded before the first MobileNetV2 run. P1 failed: the bank term was equivalent to zero in 2 of five conditions. The three inconclusive conditions are not evidence of a bank effect, since none is significant, but the prediction was equivalence and we score it as stated. P2 failed in 1 of three pairs, a reversal of the ResNet-18 result. P3 held in 5 of five conditions.

Table 14: Predictions recorded on 30 September 2026, before any MobileNetV2 run, and the outcomes.

	prediction	measured	outcome
P1	The bank term (`uniform−static`) is equivalent to zero (TOST, ± 0.5 pp) in each of the five conditions	2 of 5	failed
P2	`static` at width $2c$ is more accurate than `gap` at width c in each of the three pairs	1 of 3	failed
P3	The routing term is positive with $p < 0.05$ in at least three of the five conditions	5 of 5	held

14 What transfers between backbones

Table 15 sets the main ResNet-18 results against MobileNetV2.

Table 15: ResNet-18 against MobileNetV2, result by result. Differences are in pp and paired by seed.

result	ResNet-18	MobileNetV2	status
Bank term (`uniform − static`)	equivalent in 5 of 5	equivalent in 2 of 5; never significant (largest +0.59)	holds, weaker
Routing term significant	4 of 5	5 of 5	holds
`gap − static`, CIFAR-100 w64	+0.08 (equivalent)	+1.36 (higher)	**contradicts**
Plain $2c$ beats `gap` c (3 pairs)	3 of 3 (+4.35, +2.31, +2.52)	1 of 3 (+2.50, +0.31, +0.15)	**contradicts**
Annealed − constant τ (w16, 32, 64)	+0.11, -0.68, -0.05	-1.16, -1.28, -0.92	**differs**
`frozen_random − gap`, w64	+0.10 (both null)	+0.15; vs `uniform` +1.63	new
$K = 8$ vs $K = 2$, CIFAR-100 w32	+0.10 (equivalent)	+0.11 (inconclusive)	holds

What holds. On both backbones the bank adds nothing measurable on its own, the routing term carries the effect wherever there is one, and no bank size beyond $K = 2$ is distinguishable from $K = 2$. On MobileNetV2 static and bank kernels share one initialisation, so its bank term is free of the initialisation mismatch that affects the ResNet-18 bank term (Section 15), and it gives the same answer.

Which cost to hold fixed. The ResNet-18 conclusion that parameters are better spent on width is a statement at equal parameters. Dynamic convolution is usually advertised at equal compute. The two views disagree (Table 16). Doubling the width costs three to four times the multiply-accumulates of the routed network; routing at the same width costs at most 8 % more on ResNet-18 and 2 % more on MobileNetV2, counted for an implementation that aggregates the kernel per sample. At the same width, routing beats the plain network in 4 of five conditions on ResNet-18 and 5 of five on MobileNetV2. At equal parameters it loses all three pairs on ResNet-18 and only one of three on MobileNetV2. Whether dynamic convolution is worth its cost therefore depends on the cost and on the backbone. Our training implementation computes K convolutions per dynamic layer and is slower in wall-clock time than either count suggests (Tables 19 and 24); that is a property of the implementation, not of the layer.

The CIFAR-100 width-64 null. It does not appear on MobileNetV2. Across the ten backbone and condition pairs, the only null is ResNet-18 on CIFAR-100 at width 64, which is also the condition with the lowest static training loss (0.0080); no MobileNetV2 static arm trains below 0.248 under this recipe (Figure 15). That pattern is what the saturation account of Section 9 would expect, but ResNet-18 on Tiny-ImageNet at width 64 trains to a loss of 0.0210 and shows one of the largest effects, so no single loss threshold fits all ten points, and one chosen after seeing them would not be evidence. We do not adopt it.

Temperature and the untrained router. Annealing costs about 1 pp at every MobileNetV2 width (-1.16, -1.28, -0.92 pp at widths 16, 32 and 64), against +0.11, -0.68, -0.05 pp on ResNet-18. On MobileNetV2 the router also does not need training. Both results point the same way: what matters is that the mixture differ across images from the start of

Table 16: The routed network under two cost views. Left: `static` at width $2c$ minus `gap` at width c, with the parameter ratio routed/plain and the multiply-accumulate (MAC) ratio plain/routed. Right: `gap` minus `static` at the same width, with the MAC ratio routed/plain. MACs per image are counted for per-sample kernel aggregation.

		plain at $2c$ vs routed at c			routed vs plain, same width c	
backbone	condition	param ratio	MAC ratio	plain − routed (pp)	MAC ratio	routed − plain (pp)
ResNet-18	CIFAR-100 w16	0.99	3.7×	+4.35 (higher)	1.08×	+0.53 (higher)
ResNet-18	CIFAR-100 w32	0.99	3.7×	+2.31 (higher)	1.08×	+1.15 (higher)
ResNet-18	Tiny-ImageNet w32	0.99	3.9×	+2.52 (higher)	1.02×	+2.18 (higher)
MobileNetV2	CIFAR-100 w16	0.60	2.9×	+2.50 (higher)	1.01×	+1.91 (higher)
MobileNetV2	CIFAR-100 w32	0.51	3.1×	+0.31 (inconclusive)	1.01×	+1.31 (higher)
MobileNetV2	Tiny-ImageNet w32	0.54	3.1×	+0.15 (inconclusive)	1.00×	+1.70 (higher)

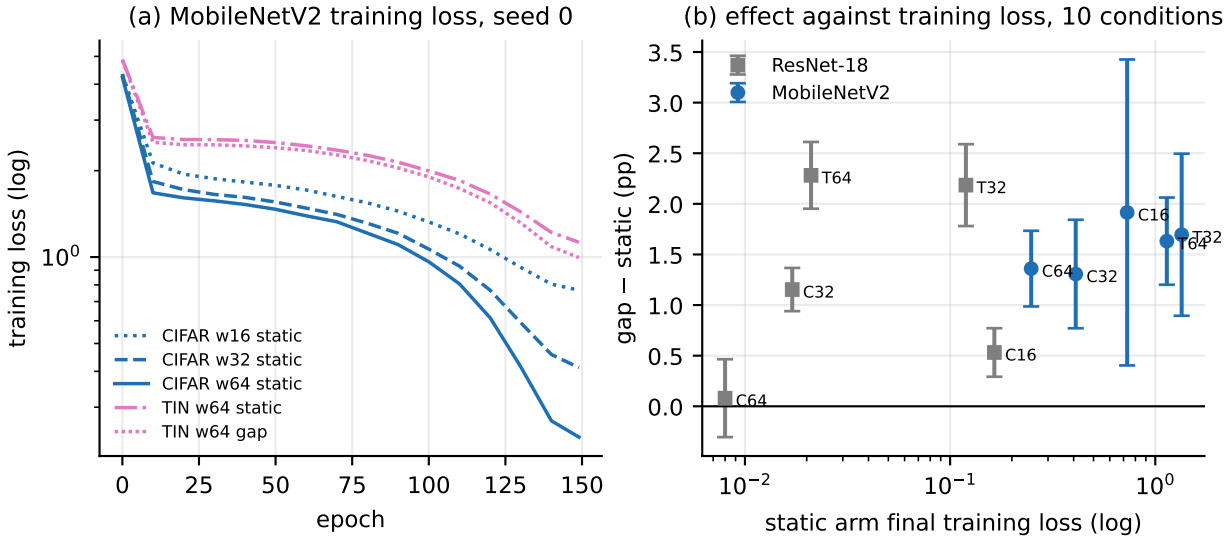


Figure 15: (a) MobileNetV2 training loss, seed 0, log scale. (b) `gap` − `static` against the static arm's final training loss for both backbones and all five conditions, 95 % CIs.

training, not that it be optimised, and a schedule that keeps it flat early removes most of the benefit. We state this as an interpretation; neither experiment was designed to test it.

Why the backbones differ. Several properties change at once between ResNet-18 and MobileNetV2: the share of parameters in dynamic layers (nearly all against 3 %), full against depthwise convolution in those layers, the size of the router relative to the bank, and how close the network comes to fitting its training set under this recipe. None was isolated here. The three remaining backbones vary these properties: VGG-16 keeps every parameter in full dynamic convolutions, ResNet-50 about half, and ConvNeXt-Tiny makes only its depthwise 7×7 layers dynamic.

15 Threats to validity

Scale. Both datasets are low-resolution and every run fits on one consumer GPU. Behaviour at ImageNet scale is untested and cannot be tested on this hardware.

Two backbones. Two backbones disagree on three of the main results, so what holds on both is the minimum that can be claimed. The same program on VGG-16, ResNet-50 and ConvNeXt-Tiny was pre-registered on 30 September 2026 and is running; its results will be added as they complete, whatever they show.

Placement and router size on MobileNetV2. Copying the ResNet-18 rule makes only the depthwise layers of MobileNetV2 dynamic, 3 % of its parameters, and its router is larger than the bank. A different rule, such as making every convolution but the stem dynamic, could change the MobileNetV2 results. The frozen-router control shows that the trainable router parameters do not produce the gain.

MobileNetV2 from the log. The MobileNetV2 numbers are taken from the training log's final lines, parsed twice by independent code with identical results. The result files on the training machine record the same values plus routing entropies; the entropy analyses (Sections 5, 11 and 12) will be repeated for MobileNetV2 from those files. One MobileNetV2 run was resumed after a GPU fault; the conclusions do not depend on it.

One recipe. Both backbones use the ResNet-18 recipe. The MobileNetV2 static arms never approach zero training loss under it, and a recipe tuned for MobileNetV2 could change where routing helps.

Initialisation. Static convolutions in the `static` arm, and the static stem, shortcut and head layers in every arm, keep PyTorch's default initialisation (Kaiming-uniform with $a = \sqrt{5}$), while every kernel of every dynamic bank is initialised Kaiming-normal with fan-out. The `uniform−static` bank term therefore compares two initialisations as well as a bank against a single kernel. The routing term, `gap−uniform`, is not affected: both arms have identically initialised banks. The replication on four more backbones initialises static and bank kernels with one scheme, and the MobileNetV2 bank term, which is free of the mismatch, also shows no bank effect.

Not a reproduction of a published system. The layer is a dynamic convolution of the CondConv and DY-Conv form. No published implementation of ODConv, FDConv or KernelWarehouse was run, and the conclusions concern dynamic convolution of this form.

Global routing. Every router here uses a global descriptor. Spatially resolved routing is not covered.

One annealing schedule. Section 11 compares a constant temperature with one linear schedule.

Stored and effective parameters. By Proposition 1 the `uniform` arm has the effective capacity of a plain network. The matched-budget comparisons of Section 10 use stored parameters. Section 14 gives the comparison at equal compute, which favours routing on both backbones.

The CIFAR-100 width-64 null is unexplained. It is well-powered and does not depend on the temperature schedule, its stated explanation failed a recorded test, and it does not appear on MobileNetV2.

Provenance and corrections. Final accuracies were transcribed from the training logs, which print them verbatim, and every figure and table is generated from those per-seed values. During the analysis, three scripts grouped results by width alone, which merged runs from the two datasets and from different bank sizes; this was detected when a comparison paired one dataset's `uniform` arm with the other's `static` arm, and every number in this report was recomputed from per-seed values keyed on dataset, width, bank size and schedule. Summary sentences printed by the analysis scripts were also found to over-read non-significant orderings on four occasions; this report relies on the per-comparison statistics only. The report generator was later rewritten to read the result files directly; on the 24 width-64 result files checked it reproduces the transcribed accuracies and routing entropies exactly. The earlier version of this report printed stored parameter counts with a third decimal that had not been measured, and six of nineteen were off by up to 0.005 M (for example 0.708 M for 0.713 M); the counts here are counted from the model definition and agree with the training logs to the two decimals they print. No comparison changes.

16 Conclusion

On two backbones, the kernel bank of dynamic convolution adds nothing measurable on its own. A uniformly mixed bank was never significantly better or worse than a plain network, and was equivalent to one in 7 of ten conditions. Whatever dynamic convolution gains, it gains through input-dependent mixing, and banks larger than $K = 2$ buy nothing further.

Whether that gain is worth having depends on the backbone and on the cost held fixed. On ResNet-18, at equal parameters, the same parameters spent on width do better by 2.3 to 4.3 pp. On MobileNetV2 the routed network is not distinguishable from a plain one with about twice its parameters in two of three pairs. At equal compute, routing helps on both. On MobileNetV2 the mixing does not have to be learned: a router frozen at random initialisation does as well as a trained one, provided each image is routed by its own descriptor.

Several results did not survive the second backbone. The ResNet-18 null on CIFAR-100 at width 64 does not appear on MobileNetV2. Temperature annealing, neutral or harmful on ResNet-18, costs about 1 pp at every MobileNetV2 width. Two of the three MobileNetV2 predictions failed, as did the recorded ResNet-18 prediction for Tiny-ImageNet. The entropy diagnostic, tested on ResNet-18, does not measure whether routing is useful.

Reproducibility. 264 runs (132 ResNet-18, 132 MobileNetV2), 458 GPU-hours, one RTX 4070 Ti. Code, the pre-registration, the dated prediction, the gate ledger with recorded evidence, and every per-seed value accompany this report.

References

S. Arora, N. Cohen, and E. Hazan. On the optimization of deep networks: Implicit acceleration by overparameterization. In *ICML*, 2018.

J. Chen, X. Wang, Z. Guo, X. Zhang, and J. Sun. Dynamic region-aware convolution. In *CVPR*, 2021.

L. Chen, L. Gu, L. Li, C. Yan, and Y. Fu. Frequency dynamic convolution for dense image prediction. In *CVPR*, pages 30178–30188, 2025.

Y. Chen, X. Dai, M. Liu, D. Chen, L. Yuan, and Z. Liu. Dynamic convolution: Attention over convolution kernels. In *CVPR*, pages 11030–11039, 2020.

X. Ding, Y. Guo, G. Ding, and J. Han. ACNet: Strengthening the kernel skeletons for powerful CNN via asymmetric convolution blocks. In *ICCV*, 2019.

X. Ding, X. Zhang, N. Ma, J. Han, G. Ding, and J. Sun. RepVGG: Making VGG-style ConvNets great again. In *CVPR*, 2021.

S. He, C. Jiang, D. Dong, and L. Ding. SD-Conv: Towards the parameter-efficiency of dynamic convolution. In *WACV*, 2023.

K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *CVPR*, 2016.

A. Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

D. Lakens. Equivalence tests: A practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science*, 8(4):355–362, 2017.

Y. Le and X. Yang. Tiny ImageNet visual recognition challenge. CS231N course report, Stanford University, 2015.

C. Li and A. Yao. KernelWarehouse: Rethinking the design of dynamic convolution. In *ICML*, PMLR 235:29201–29221, 2024.

C. Li, A. Zhou, and A. Yao. Omni-dimensional dynamic convolution. In *ICLR*, 2022.

Y. Li, Y. Chen, X. Dai, M. Liu, D. Chen, Y. Yu, L. Yuan, Z. Liu, M. Chen, and N. Vasconcelos. Revisiting dynamic convolution via matrix decomposition. In *ICLR*, 2021.

M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *CVPR*, 2018.

D. J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680, 1987.

F. Wilcoxon. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83, 1945.

B. Yang, G. Bender, Q. V. Le, and J. Ngiam. CondConv: Conditionally parameterized convolutions for efficient inference. In *NeurIPS*, 2019.

C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals. Understanding deep learning requires rethinking generalization. In *ICLR*, 2017.

A Complete results: ResNet-18

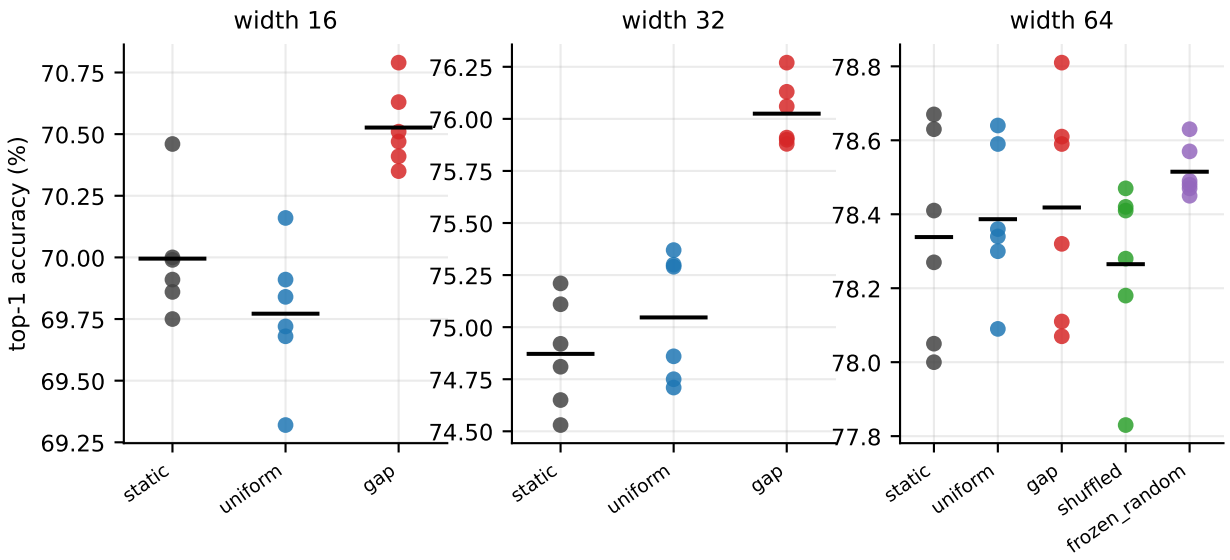


Figure 16: Every CIFAR-100 run at $K = 4$ and $\tau = 1$; bars are arm means.

Table 17: Every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%) for each seed, with mean and sample standard deviation.

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	69.91	69.75	69.86	70.46	70.00	69.99	69.995	0.246
CIFAR-100	16	`uniform`	69.68	69.32	69.84	69.91	69.72	70.16	69.772	0.279
CIFAR-100	16	`gap`	70.35	70.51	70.41	70.63	70.79	70.47	70.527	0.160
CIFAR-100	32	`static`	74.81	74.65	74.53	74.92	75.11	75.21	74.872	0.262
CIFAR-100	32	`uniform`	75.29	74.86	74.75	75.30	74.71	75.37	75.047	0.305
CIFAR-100	32	`gap`	76.13	75.88	75.91	75.90	76.27	76.06	76.025	0.156
CIFAR-100	64	`static`	78.05	78.67	78.63	78.27	78.00	78.41	78.338	0.284
CIFAR-100	64	`uniform`	78.64	78.59	78.34	78.30	78.36	78.09	78.387	0.202
CIFAR-100	64	`gap`	78.59	78.81	78.11	78.61	78.07	78.32	78.418	0.299
CIFAR-100	64	`shuffled`	78.42	77.83	78.18	78.41	78.47	78.28	78.265	0.238
CIFAR-100	64	`frozen_random`	78.47	78.49	78.48	78.63	78.45	78.57	78.515	0.070
Tiny-ImageNet	32	`static`	61.93	61.61	61.57	61.46	61.27	61.52	61.560	0.217
Tiny-ImageNet	32	`uniform`	61.52	61.53	61.55	61.62	61.66	62.12	61.667	0.229
Tiny-ImageNet	32	`gap`	63.53	64.20	63.61	63.58	63.91	63.64	63.745	0.260
Tiny-ImageNet	64	`static`	66.29	66.59	65.88	65.97	66.63	66.23	66.265	0.309
Tiny-ImageNet	64	`uniform`	66.27	66.27	66.41	66.21	66.22	66.53	66.318	0.126
Tiny-ImageNet	64	`gap`	68.73	68.35	68.42	68.55	68.71	68.52	68.547	0.152

Table 18: Paired comparisons against `static`, $n = 6$. p_W is the Wilcoxon signed-rank test, which cannot fall below 0.031 at this n.

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	-0.223	[-0.500, +0.054]	0.0931	0.0938	-0.85	0.0251	equivalent
CIFAR-100	16	`gap`	+0.532	[+0.292, +0.771]	0.0023	0.0312	+2.33	0.6260	higher
CIFAR-100	32	`uniform`	+0.175	[-0.146, +0.496]	0.2206	0.3125	+0.57	0.0242	equivalent
CIFAR-100	32	`gap`	+1.153	[+0.940, +1.367]	$< 10^{-4}$	0.0312	+5.67	0.9997	higher
CIFAR-100	64	`uniform`	+0.048	[-0.332, +0.429]	0.7573	0.8438	+0.13	0.0142	equivalent
CIFAR-100	64	`gap`	+0.080	[-0.305, +0.465]	0.6158	0.5625	+0.22	0.0188	equivalent
CIFAR-100	64	`shuffled`	-0.073	[-0.602, +0.456]	0.7361	1.0000	-0.15	0.0464	equivalent
CIFAR-100	64	`frozen_random`	+0.177	[-0.121, +0.474]	0.1874	0.2188	+0.62	0.0191	equivalent
Tiny-ImageNet	32	`uniform`	+0.107	[-0.270, +0.483]	0.4994	0.6875	+0.30	0.0218	equivalent
Tiny-ImageNet	32	`gap`	+2.185	[+1.781, +2.589]	$< 10^{-4}$	0.0312	+5.67	0.9999	higher
Tiny-ImageNet	64	`uniform`	+0.053	[-0.334, +0.441]	0.7380	1.0000	+0.14	0.0157	equivalent
Tiny-ImageNet	64	`gap`	+2.282	[+1.952, +2.611]	$< 10^{-4}$	0.0312	+7.26	1.0000	higher

Table 19: Configuration and cost. Router parameters are in thousands and are included in the stored total. Training loss is the final-epoch mean over six seeds.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.71	0.0	0.1643	9.6
CIFAR-100	16	`uniform`	2.77	0.0	0.1653	11.4
CIFAR-100	16	`gap`	2.79	21.3	0.0638	33.7
CIFAR-100	32	`static`	2.82	0.0	0.0170	8.6
CIFAR-100	32	`uniform`	11.06	0.0	0.0170	12.9
CIFAR-100	32	`gap`	11.13	74.6	0.0132	39.2
CIFAR-100	64	`static`	11.22	0.0	0.0080	18.8
CIFAR-100	64	`uniform`	44.18	0.0	0.0081	25.2
CIFAR-100	64	`gap`	44.46	288.0	0.0074	86.0
CIFAR-100	64	`shuffled`	44.46	288.0	0.0083	92.3
CIFAR-100	64	`frozen_random`	44.18	0.0	0.0077	85.4
Tiny-ImageNet	32	`static`	2.85	0.0	0.1194	54.2
Tiny-ImageNet	32	`uniform`	11.09	0.0	0.1200	56.3
Tiny-ImageNet	32	`gap`	11.16	74.6	0.0420	284.2
Tiny-ImageNet	64	`static`	11.27	0.0	0.0210	129.0
Tiny-ImageNet	64	`uniform`	44.23	0.0	0.0220	141.0
Tiny-ImageNet	64	`gap`	44.52	288.0	0.0190	687.2

Table 20: Routing entropy by arm, CIFAR-100 width 64, per-seed means.

arm	mean $\tilde{H}$	min	max	$1 - \tilde{H}$
`uniform`	1.0000	1.0000	1.0000	0.00000
`gap`	0.9885	0.9867	0.9912	0.01147
`shuffled`	0.9996	0.9995	0.9997	0.00040
`frozen_random`	0.9971	0.9967	0.9977	0.00287

Table 21: Every run in the study. GPU-hours are from logged wall-clock time.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	5.5
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	6.1
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	30.8
CIFAR-100	32	2	constant	`gap`	6	2.8
CIFAR-100	32	8	constant	`gap`	6	6.9
CIFAR-100	16	4	annealed	`gap`	6	4.4
CIFAR-100	32	4	annealed	`gap`	6	3.8
CIFAR-100	64	4	annealed	`gap`	6	8.6
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	39.5
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	95.7
total					132	204.1

B Complete results: MobileNetV2

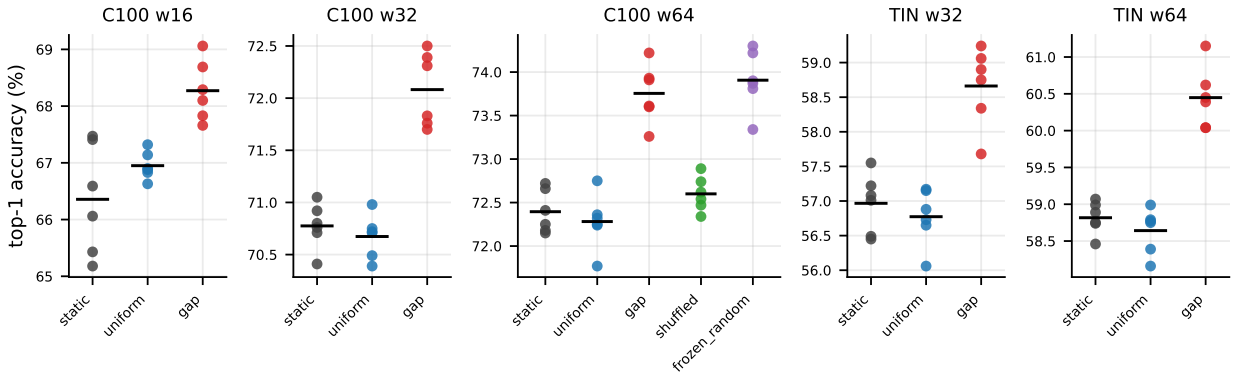


Figure 17: Every MobileNetV2 run at $K = 4$ and $\tau = 1$; bars are arm means.

Table 22: MobileNetV2, every run at $K = 4$ and $\tau = 1$: final-epoch top-1 accuracy (%). † marks the run resumed after a GPU fault.

dataset	width	arm	s0	s1	s2	s3	s4	s5	mean	sd
CIFAR-100	16	`static`	65.43	67.41	66.59	65.18	67.47	66.06	66.357	0.973
CIFAR-100	16	`uniform`	66.88	66.90	66.63	67.14	67.32	66.83	66.950	0.244
CIFAR-100	16	`gap`	68.69	67.83	67.66	69.06	68.10	68.29	68.272	0.528
CIFAR-100	32	`static`	70.92	71.05	70.71	70.80	70.41	70.76	70.775	0.217
CIFAR-100	32	`uniform`	70.49	70.39	70.98	70.72	70.75	70.71	70.673	0.208
CIFAR-100	32	`gap`	72.39	71.76	72.31	71.70	72.50	71.83	72.082	0.356
CIFAR-100	64	`static`	72.66	72.18	72.15	72.25	72.41	72.72	72.395	0.246
CIFAR-100	64	`uniform`	72.32	71.77	72.75	72.24	72.25	72.36	72.282	0.314
CIFAR-100	64	`gap`	74.22	73.26	73.93	73.91	73.60	73.61	73.755	0.335
CIFAR-100	64	`shuffled`	72.34	72.62	72.54	72.74	72.89	72.47	72.600	0.196
CIFAR-100	64	`frozen_random`	74.22	74.30	73.81	73.34	73.87	73.90	73.907	0.342
Tiny-ImageNet	32	`static`	57.01	57.08	56.45	57.22	56.49	57.55	56.967	0.427
Tiny-ImageNet	32	`uniform`	56.06	56.65	56.88	56.73	57.17	57.15	56.773	0.409
Tiny-ImageNet	32	`gap`	58.90	58.34	59.06	57.68	58.75	59.24	58.662	0.570
Tiny-ImageNet	64	`static`	59.07	58.99	58.89	58.74	58.75	58.46	58.817	0.218
Tiny-ImageNet	64	`uniform`	58.39	58.16	58.99	58.75	58.77	58.79	58.642	0.306
Tiny-ImageNet	64	`gap`	60.45	61.15†	60.04	60.62	60.04	60.39	60.448	0.414

Table 23: MobileNetV2: paired comparisons against `static`, $n = 6$.

dataset	width	arm	Δ (pp)	95 % CI	p_t	p_W	d_z	p_{TOST}	verdict
CIFAR-100	16	`uniform`	+0.593	[-0.425, +1.612]	0.1945	0.3125	+0.61	0.5884	inconclusive
CIFAR-100	16	`gap`	+1.915	[+0.404, +3.426]	0.0225	0.0312	+1.33	0.9694	higher
CIFAR-100	32	`uniform`	-0.102	[-0.510, +0.306]	0.5501	0.5625	-0.26	0.0269	equivalent
CIFAR-100	32	`gap`	+1.307	[+0.771, +1.843]	0.0015	0.0312	+2.56	0.9941	higher
CIFAR-100	64	`uniform`	-0.113	[-0.512, +0.285]	0.4978	0.4375	-0.30	0.0275	equivalent
CIFAR-100	64	`gap`	+1.360	[+0.986, +1.734]	0.0002	0.0312	+3.82	0.9990	higher
CIFAR-100	64	`shuffled`	+0.205	[-0.196, +0.606]	0.2456	0.1562	+0.54	0.0585	inconclusive
CIFAR-100	64	`frozen_random`	+1.512	[+1.123, +1.900]	0.0002	0.0312	+4.08	0.9994	higher
Tiny-ImageNet	32	`uniform`	-0.193	[-0.842, +0.455]	0.4782	0.6250	-0.31	0.1393	inconclusive
Tiny-ImageNet	32	`gap`	+1.695	[+0.895, +2.495]	0.0028	0.0312	+2.22	0.9939	higher
Tiny-ImageNet	64	`uniform`	-0.175	[-0.664, +0.314]	0.4001	1.0000	-0.38	0.0742	inconclusive
Tiny-ImageNet	64	`gap`	+1.632	[+1.201, +2.062]	0.0002	0.0312	+3.98	0.9995	higher

Table 24: MobileNetV2: configuration and cost. Parameters are counted from the model definition; router parameters are in thousands (the frozen router is held as buffers and not counted). Training loss is the final-epoch mean over six seeds; minutes are logged wall-clock time.

dataset	width	arm	params (M)	router (K)	train loss	min/run
CIFAR-100	16	`static`	0.37	0.0	0.729	16.2
CIFAR-100	16	`uniform`	0.42	0.0	0.717	20.2
CIFAR-100	16	`gap`	0.49	75.3	0.655	41.0
CIFAR-100	32	`static`	0.82	0.0	0.409	18.0
CIFAR-100	32	`uniform`	0.91	0.0	0.418	21.6
CIFAR-100	32	`gap`	1.20	289.1	0.344	47.3
CIFAR-100	64	`static`	2.35	0.0	0.248	24.8
CIFAR-100	64	`uniform`	2.54	0.0	0.231	26.9
CIFAR-100	64	`gap`	3.68	1137.3	0.168	78.1
CIFAR-100	64	`shuffled`	3.68	1137.3	0.229	92.0
CIFAR-100	64	`frozen_random`	2.54	0.0	0.175	77.4
Tiny-ImageNet	32	`static`	0.94	0.0	1.345	116.1
Tiny-ImageNet	32	`uniform`	1.04	0.0	1.359	114.5
Tiny-ImageNet	32	`gap`	1.33	289.1	1.220	340.6
Tiny-ImageNet	64	`static`	2.48	0.0	1.138	187.4
Tiny-ImageNet	64	`uniform`	2.67	0.0	1.126	191.3
Tiny-ImageNet	64	`gap`	3.81	1137.3	0.995	792.5

Table 25: Every MobileNetV2 run. GPU-hours are from logged wall-clock time.

dataset	width	K	τ	arms	runs	GPU-h
CIFAR-100	16	4	constant	`static`, `uniform`, `gap`	18	7.7
CIFAR-100	32	4	constant	`static`, `uniform`, `gap`	18	8.7
CIFAR-100	64	4	constant	`static`, `uniform`, `gap`, `shuffled`, `frozen_random`	30	29.9
CIFAR-100	32	2	constant	`gap`	6	5.2
CIFAR-100	32	8	constant	`gap`	6	9.9
CIFAR-100	16	4	annealed	`gap`	6	4.1
CIFAR-100	32	4	annealed	`gap`	6	4.7
CIFAR-100	64	4	annealed	`gap`	6	9.1
Tiny-ImageNet	32	4	constant	`static`, `uniform`, `gap`	18	57.1
Tiny-ImageNet	64	4	constant	`static`, `uniform`, `gap`	18	117.1
total					132	253.6